SSL-KD: A Self-Supervised Teacher–Student Framework for Lightweight ECG Classification

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***Abstract*— Electrocardiogram (ECG) classification has become an important application of deep learning for the early diagnosis of cardiovascular diseases. While recent deep learning models achieve high diagnostic accuracy, they often require large annotated datasets and they are computationally intensive architectures, limiting their deployment in real-time and resource-constrained environments. Lightweight models based on knowledge distillation improve efficiency but still rely heavily on supervised learning. In this work, we propose a Self-Supervised Knowledge Distillation (SSL-KD) framework for efficient ECG classification. The proposed approach first learns robust ECG representations through self-supervised pretraining on unlabeled signals, followed by supervised fine-tuning of a teacher network. The learned knowledge is then transferred to a lightweight student model using knowledge distillation. This enables fast inference while preserving important feature representations. The proposed framework aims to reduce dependence on labeled data and improve the performance of compact ECG classification models suitable for edge and wearable healthcare applications.**

I. Introduction

Cardiovascular diseases (CVDs) are one of the major causes of death worldwide [1]. Early detection of these diseases can help doctors provide proper treatment and improve patient survival. Electrocardiogram (ECG) is one of the most widely used diagnostic tools for monitoring the electrical activity of the heart. It is non-invasive, portable, inexpensive, and easy to perform. Because of these advantages, ECG is commonly used in hospitals, emergency care, ambulances, and wearable healthcare devices.

Although ECG recording is simple, interpreting it correctly is not easy. A reliable diagnosis usually requires an experienced cardiologist or trained medical expert. Manual ECG interpretation is time-consuming and may vary from one expert to another. In many rural and remote areas, experienced cardiologists are not always available. This has motivated researchers to develop Artificial Intelligence (AI) models that can automatically analyze ECG signals and assist doctors in disease diagnosis.

Earlier ECG classification methods mainly used handcrafted features with machine learning algorithms. These methods required domain knowledge to design useful features, and their performance depended on the quality of those features. Later, deep learning methods became popular because they can learn useful features directly from raw ECG signals without manual feature engineering. Different architectures based on Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and attention mechanisms have shown significant improvements in ECG classification performance.

One such model is IMLE-Net [2], which learns ECG representations at multiple levels, namely beat level, rhythm level, and lead level. By combining these hierarchical features, the network captures both local and global information from all twelve ECG leads. It also provides attention maps that help explain which parts of the ECG contribute to the final prediction. Although IMLE-Net achieves high classification accuracy, its hierarchical architecture contains multiple processing stages, resulting in a large model with higher computational cost and longer inference time. This makes it difficult to deploy in real-time applications and resource-constrained edge devices.

To reduce the computational complexity, LRH-Net [3] introduces a lightweight ECG classification model based on a teacher-student framework. It uses Multi-Level Knowledge Distillation (MLKD) to transfer knowledge from a large teacher model to a compact student model. This significantly reduces the number of parameters, memory usage, and inference time while maintaining competitive classification performance. However, LRH-Net is completely based on supervised learning, which requires a large amount of accurately labeled ECG data. Obtaining these labels requires expert cardiologists, making the annotation process expensive and time-consuming.

Recently, self-supervised learning has shown promising results in learning useful feature representations from unlabeled data. Instead of depending only on manually annotated samples, self-supervised learning allows the model to learn meaningful ECG representations from a large collection of unlabeled signals. However, very few studies have combined self-supervised learning with lightweight ECG classification and knowledge distillation for efficient deployment.

In this work, we propose a Self-Supervised Knowledge Distillation (SSL-KD) framework for ECG classification. The proposed framework first performs self-supervised pretraining on unlabeled ECG signals to learn robust feature representations. The pretrained teacher network is then fine-tuned using labeled ECG data for disease classification. Finally, the knowledge learned by the teacher is transferred to a lightweight student network through knowledge distillation. The proposed framework aims to reduce the dependence on labeled ECG data while maintaining a compact model with fast inference, making it suitable for real-time and edge-based healthcare applications.

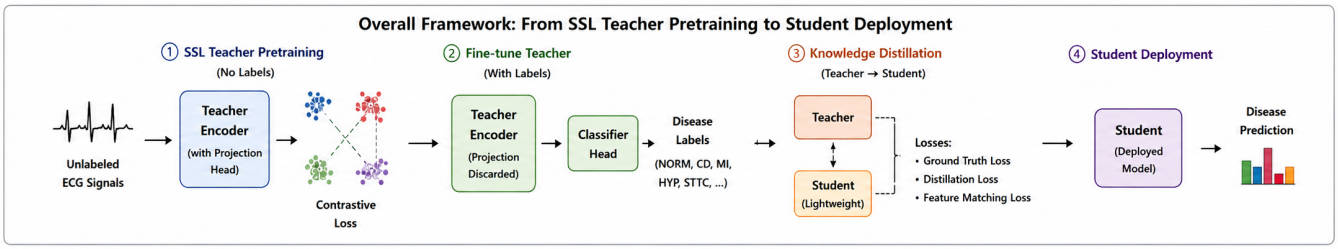


Fig. 1. Overview of the proposed self-supervised learning and knowledge distillation framework.

II. Methodology

A. Architecture

The proposed SSL-KD framework consists of three sequential stages: self-supervised pretraining of a teacher encoder, supervised fine-tuning of the pretrained teacher, and knowledge distillation [4] to a lightweight student network. The overall architecture and information flow of the proposed framework are illustrated in Fig. 1.

The framework adopts a dual-branch architecture to capture complementary rhythm and morphological information from the 12-lead ECG. Lead II is provided to the rhythm branch, while the remaining eleven leads are processed by the morphology branch. For the teacher, both branches operate on the complete preprocessed ECG sequence. The teacher network employs residual convolutional blocks [5], squeeze-and-excitation modules [6], temporal sequence modeling, and attention-based feature extraction [7]. The representations learned by the two branches are integrated through a cross-attention-based fusion mechanism, producing a 512-dimensional ECG representation.

During self-supervised pretraining, the teacher encoder is trained using unlabeled MIMIC-IV-ECG recordings. Two independently augmented views of the same ECG are generated and processed by the dual-branch encoder. The resulting representations are optimized using a symmetrized SimSiam [8] objective together with an auxiliary cross-branch consistency objective. The SimSiam formulation uses stop-gradient on the target representations, allowing the encoder to learn discriminative ECG representations without requiring diagnostic annotations.

Following self-supervised pretraining, the complete teacher encoder is fine-tuned using labeled CinC2020 recordings. Patient metadata consisting of age and gender is incorporated alongside the learned ECG representation. The metadata is projected from a 5-dimensional input representation to a 10-dimensional embedding and concatenated with the 512-dimensional ECG representation, resulting in a 522-dimensional teacher embedding. A linear classification layer then maps this representation to the 24 diagnostic classes.

The final stage transfers the learned knowledge of the teacher to a lightweight student network. The student retains the dual-branch design but uses substantially reduced computational components, including a bidirectional GRU [9] for rhythm modeling and depthwise separable convolutions [10] for morphological feature extraction. The rhythm branch processes the complete 4096-sample ECG sequence, whereas the morphology branch operates on a randomly selected 642-sample temporal segment. The two student branches are fused to produce a compact 64-dimensional representation. Unlike the teacher, the student does not directly use patient metadata.

To enable feature-level knowledge transfer, a linear projection maps the 64-dimensional student representation into the 522-dimensional teacher embedding space. Knowledge is transferred at both representation and prediction levels. Specifically, the student is optimized using supervised classification loss, feature-level distillation loss, and logit-level distillation loss, while the teacher parameters remain frozen during distillation.

The teacher contains 7,164,116 trainable parameters, whereas the student contains only 104,834 parameters. Thus, the proposed student achieves a $68.34\times$ reduction in the number of parameters relative to the teacher, substantially reducing the model size while retaining the dual-branch ECG processing and knowledge transfer mechanisms.

B. Preprocessing

Two ECG datasets are used in this work, with separate preprocessing pipelines to accommodate their different sampling characteristics.

The **MIMIC-IV-ECG** [11] dataset is the Diagnostic Electrocardiogram Matched Subset available through PhysioNet [12]. The recordings are processed online during self-supervised training. Missing values are replaced with zeros using `np.nan_to_num(·)`, and the signals are resampled to 250 Hz using piecewise linear interpolation. No filtering, normalization, or amplitude clipping is applied. Following resampling, the implementation retains up to 3750 samples per lead, corresponding to a maximum temporal duration of 15 s at 250 Hz.

The **PhysioNet/CinC 2020** [13] dataset contains multi-source twelve-lead ECG recordings with varying original sampling frequencies. The preprocessing pipeline first resamples each recording to 257 Hz using piecewise linear interpolation. A third-order zero-phase Butterworth bandpass filter is then applied. The filtered signals undergo per-lead normalization, followed by numerical handling of missing values and a second mean-standard-deviation normalization. Finally, recordings shorter than 4096 samples are symmetrically zero-padded, whereas longer recordings are truncated to 4096 samples.

After preprocessing, the twelve-lead ECG is separated into rhythm and morphology branches. Lead II (lead index 1) is used as the rhythm input, while the remaining eleven leads are used as the morphology input. For CinC 2020, the teacher therefore receives inputs of dimensions $(1, 4096)$ and $(11, 4096)$ for the rhythm and morphology branches, respectively.

During knowledge distillation, the student uses the same preprocessed CinC 2020 recordings but employs a shorter temporal crop for the morphology branch. The rhythm branch retains the complete 4096-sample sequence, while the morphology branch is randomly cropped to 642 samples, corresponding to approximately 2.5 s at 257 Hz. This asymmetric temporal input allows the student model to operate with substantially lower computational and parameter requirements while retaining the full rhythm information.

C. Self-Supervised Pretraining

To learn informative ECG representations without relying on diagnostic labels, a self-supervised pretraining stage is performed using the MIMIC-IV-ECG dataset. A dual-branch teacher encoder is employed to separately capture rhythm and morphological characteristics of the ECG. The overall teacher architecture is shown in Fig. 2.

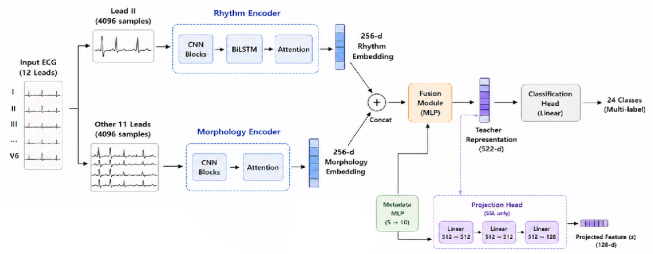


Fig. 2. Architecture of the self-supervised teacher encoder. Lead II is processed by the rhythm branch, while the remaining 11 leads are processed by the morphology branch. The branch representations are fused with demographic metadata to obtain the 522-dimensional teacher representation.

The rhythm branch processes Lead II using convolutional blocks followed by a bidirectional long short-term memory (BiLSTM) [14] layer and a temporal attention mechanism [7], producing a 512-dimensional rhythm representation. The remaining eleven leads are processed by the morphology branch using multi-lead convolutional residual blocks with squeeze-and-excitation modules and temporal attention, producing a 256-dimensional morphology representation. The morphology representation is first projected from 256 to 512 dimensions to match the rhythm representation. A cross-attention module then integrates the rhythm and morphology features. The resulting attended rhythm and normalized morphology representations are concatenated, forming a 1024-dimensional feature vector, which is subsequently processed by a fusion MLP to obtain the final 512-dimensional ECG representation.

Patient metadata is represented using a five-dimensional vector containing the available age and gender information. A small multilayer perceptron projects this metadata to 10 dimensions, which is combined with the 512-dimensional ECG feature to form the final 522-dimensional teacher representation. During supervised fine-tuning, this representation is used by the 24-class multi-label classification head.

During self-supervised pretraining, the teacher encoder is trained using a symmetrized SimSiam objective. Two independently augmented views are generated from the same ECG recording and passed through the shared teacher encoder. The resulting representations are processed by a projection head and predictor, as illustrated in Fig. 3.

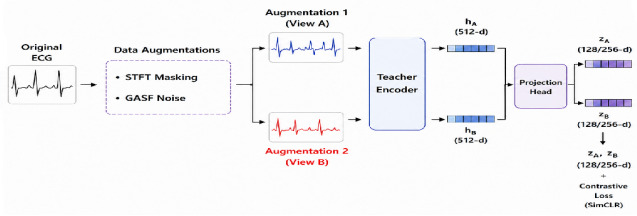


Fig. 3. Self-supervised pretraining framework. Two independently augmented views of the same ECG are processed by the shared teacher encoder, followed by projection and prediction heads for the SimSiam objective.

For each ECG, two views are generated independently using STFT masking and GASF-based noise augmentation. The same encoder parameters are shared between the two views. The projection head maps the 512-dimensional fused representation to a 128-dimensional projection, while the predictor maps this projection through a bottleneck and back to a 128-dimensional representation.

The self-supervised objective consists of a symmetric fused representation loss and an auxiliary cross-branch loss. Let $D(\cdot, \cdot)$ denote the negative cosine similarity. The fused loss is defined as

$$\mathcal{L}_{\text{fused}} = \frac{1}{2} D(p_1, z_2) + \frac{1}{2} D(p_2, z_1), \tag{1}$$

where z_1 and z_2 are stop-gradient projected representations of the two augmented views, and p_1 and p_2 are their corresponding predictions.

An additional cross-branch objective encourages consistency between rhythm and morphology representations:

$$\mathcal{L}_{\text{cross}} = \frac{1}{2} D(p_{1,r}, z_{1,m}) + \frac{1}{2} D(p_{1,m}, z_{1,r}), \tag{2}$$

where the subscripts r and m denote the rhythm and morphology branches, respectively. The total self-supervised loss is

$$\mathcal{L}_{\text{SSL}} = \mathcal{L}_{\text{fused}} + 0.5\, \mathcal{L}_{\text{cross}}. \tag{3}$$

The teacher is pretrained for 120 epochs using the Adam optimizer [15] with a learning rate of 3×10^{-4} and a weight decay of 10^{-5}. A batch size of 32 is used, with mixed-precision training enabled. The projection and predictor heads are used only during self-supervised pretraining and are discarded during subsequent teacher fine-tuning and knowledge distillation.

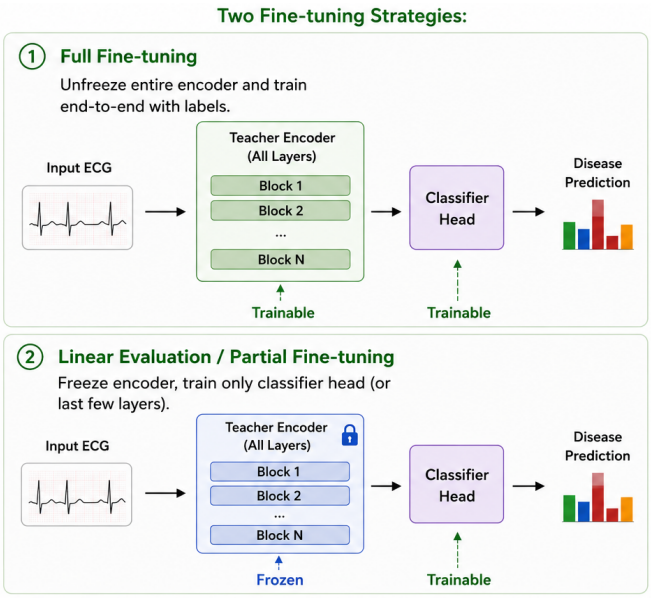


Fig. 4. Teacher fine-tuning framework. The pretrained teacher encoder is jointly optimized with a classification head for multi-label ECG disease prediction.

D. Teacher Fine-Tuning

Following self-supervised pretraining, the pretrained teacher encoder is fine-tuned on the PhysioNet/CinC 2020 dataset for multi-label ECG classification, as illustrated in Fig. 4. The projection and predictor heads used during self-supervised pretraining are discarded, while the pretrained encoder is retained as the feature extractor.

Full fine-tuning is used in this work, in which all parameters of the pretrained teacher encoder are updated together with the newly initialized classification head. This strategy is adopted because the self-supervised pretraining stage learns general ECG representations without using diagnostic labels, whereas the supervised CinC2020 dataset provides task-specific diagnostic information. Updating the complete encoder therefore allows the learned representations to adapt to the diagnostic characteristics of the target dataset rather than keeping the pretrained features fixed.

The teacher encoder produces a 512-dimensional ECG representation. Patient metadata used by the teacher is projected to a 10-dimensional representation and concatenated with this 512-dimensional ECG representation, resulting in a 522-dimensional fused representation, which is passed to a linear classification layer. The classification head produces 24 output logits corresponding to the diagnostic classes defined for the CinC2020 dataset.

The model is trained for 90 epochs using the AdamW optimizer [16] with differential learning rates of 3×10^{-4} for the pretrained encoder and 3×10^{-3} for the classification head. A weight decay of 10^{-4} is applied during optimization. Asymmetric Loss [17] is used for multi-label classification to account for the imbalance among the diagnostic classes. No label smoothing, gradient clipping, learning-rate scheduler, or mixed-precision training is used during teacher fine-tuning.

The resulting teacher model contains 7,164,116 parameters, including 6,993,468 parameters in the encoder and 170,648 parameters in the classification and metadata-fusion components. The fine-tuned teacher is subsequently used as the teacher network for knowledge distillation into the lightweight student model.

E. Knowledge Distillation

After supervised fine-tuning, the teacher model is used to transfer its learned representations and diagnostic knowledge to a lightweight student network, as illustrated in Fig. 5. The objective is to obtain a substantially smaller model while retaining the diagnostic information learned by the teacher.

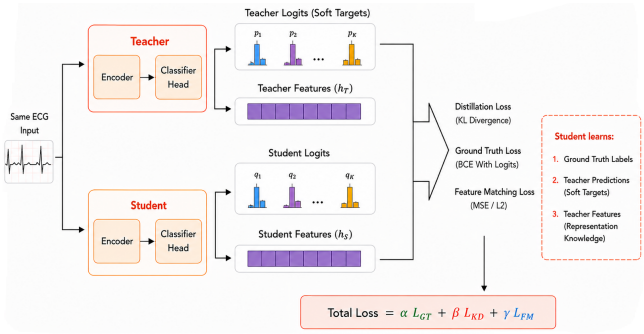


Fig. 5. Knowledge distillation framework. The fine-tuned teacher transfers feature-level and logit-level knowledge to the lightweight student network.

During distillation, the teacher parameters are frozen and the teacher is kept in evaluation mode. The teacher produces a 522-dimensional fused representation, which contains the 512-dimensional ECG representation combined with the 10-dimensional metadata representation. The student network produces a compact 64-dimensional representation, which is projected to the same 522-dimensional space using a linear projection layer before feature-level distillation.

The student is trained using three complementary objectives: supervised classification loss, feature-level distillation loss, and logit-level distillation loss. The supervised classification component uses Asymmetric Focal Loss [17] with $\gamma_{\text{neg}} = 4$, $\gamma_{\text{pos}} = 1$, and a probability clipping value of 0.05. The feature-level distillation loss is defined using the mean squared error between the teacher representation and the projected student representation:

$$\mathcal{L}_{\text{feat}} = \left\| h_T - \hat{h}_S \right\|_2^2, \tag{4}$$

where $h_T \in R^{522}$ denotes the teacher representation and $\hat{h}_S \in R^{522}$ denotes the projected student representation.

To transfer the teacher's output-level knowledge, the student and teacher logits are compared using the Kullback–Leibler divergence with a temperature of $T = 2$. The complete distillation objective is given by

$$\mathcal{L}_{\text{KD}} = \alpha \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{feat}} + \gamma T^2 \mathcal{L}_{\text{KL}}, \tag{5}$$

where $\mathcal{L}_{\text{cls}}$ is the supervised classification loss, $\mathcal{L}_{\text{feat}}$ is the feature distillation loss, and $\mathcal{L}_{\text{KL}}$ is the logit distillation loss. The weighting coefficients are set to $\alpha = 0.3$, $\beta = 0.6$, and $\gamma = 0.1$.

The student is optimized using Adam with a learning rate of 0.001 for 80 epochs and a batch size of 64. The final student model contains 104,834 parameters compared with 7,164,116 parameters in the teacher, corresponding to a parameter reduction of approximately $68.34\times$. Despite this substantial reduction, the student retains the same 24-dimensional output space as the teacher for multi-label diagnostic prediction.

III. Experiments

A. Experimental Setup

All experiments were conducted using the PhysioNet/CinC 2020 dataset [13] for supervised teacher fine-tuning and student knowledge distillation, evaluated using the predefined training/validation split specified by the corresponding metadata files. The self-supervised teacher pretraining was performed separately on the unlabeled MIMIC-IV-ECG dataset [11]. The optimizer, learning rate, weight decay, batch size, and number of epochs used at each of the three training stages – self-supervised pretraining, teacher fine-tuning, and knowledge distillation – follow the configuration described in Section II and are not repeated here.

The teacher contains 7,164,116 trainable parameters, whereas the proposed student contains 104,834 parameters, corresponding to a parameter reduction of approximately $68.34\times$. The teacher produces a 522-dimensional fused representation, while the student uses a 64-dimensional internal representation followed by a projection layer to match the teacher representation for feature-level distillation.

B. Evaluation Metrics

The teacher and student models are evaluated using Macro AUROC, Micro AUROC, Macro F1-score, Micro F1-score, Fmax, and the PhysioNet Challenge metric (CM Score). Macro AUROC and Macro F1 compute the unweighted average across the 24 diagnostic classes, whereas Micro AUROC and Micro F1 aggregate the predictions across all classes. For threshold-based metrics, a threshold of 0.5 is used to obtain binary predictions. Fmax is obtained by evaluating Macro F1 over thresholds from 0.01 to 0.99, in increments of 0.01, and selecting the maximum value. The CM Score follows the official PhysioNet/CinC 2020 challenge formulation and uses the challenge weight matrix. Per-class AUROC, precision, recall, and F1-score are additionally reported to analyze class-specific performance.

C. Overall Performance

Table I summarizes the performance of the final teacher and student models on the validation set. The teacher achieves a CM Score of 0.6527, a Macro AUROC of 0.9486, and a Macro F1 of 0.6033. The student achieves a CM Score of 0.5378, a Macro AUROC of 0.8949, and a Macro F1 of 0.4419. Despite having substantially fewer parameters, the student retains a considerable portion of the teacher's classification performance.

TABLE I

Performance of the final teacher and student models.

Model	CM Score	Macro AUROC	Macro F1
Teacher	0.6527	0.9486	0.6119
Student	0.5378	0.8949	0.4419

Table II provides the detailed validation metrics for the final models. In addition to the metrics reported above, the student achieves a Micro AUROC of 0.9511 and a Micro F1 of 0.7220. These results provide a more detailed view of the multi-label classification performance of the distilled student model.

TABLE II

Validation performance of the proposed teacher and student models.

Model	CM Score	Macro AUROC	Micro AUROC	Macro F1	Micro F1
Teacher	0.6527	0.9486	0.9769	0.6119	0.7536
Student	0.5378	0.8949	0.9511	0.4419	0.7220

D. Comparison with Baseline Models

Table III compares the proposed teacher and student models with reproduced baseline models evaluated on the PhysioNet/CinC 2020 dataset. The reproduced baselines include SE-ResNet with 12 leads, SE-ResNet with 2 leads, 1D-CNN with 2 leads, and LRH-Net [3] with 2 leads. The comparison is based on the reproduced CM Score and Micro F1 values.

TABLE III

Comparison with reproduced baseline models on the PhysioNet/CinC 2020 dataset.

Model	CM Score	Micro F1
SE-ResNet (12-Lead)	0.6673	0.7549
SE-ResNet (2-Lead)	0.6286	0.7417
1D-CNN (2-Lead)	0.5684	0.6966
LRH-Net (2-Lead)	0.4923	0.6604
Ours – Teacher (12-Lead)	0.6527	0.7536
Ours – Student (1-Lead)	0.5378	0.7220

The proposed teacher achieves a CM Score of 0.6527, while the proposed student achieves 0.5378. The student obtains a Micro F1 of 0.7220, which is higher than the reproduced LRH-Net and 1D-CNN baselines shown in the table.

E. Comparison with IMLE-Net

Table IV provides a contextual comparison between the reproduced IMLE-Net results and the proposed student architecture using the corresponding evaluation metrics. IMLE-Net was originally evaluated on the PTB-XL dataset, whereas the proposed student model was evaluated on the PhysioNet/CinC 2020 dataset. Therefore, the reported values should not be interpreted as a direct quantitative comparison, since the datasets, label distributions, and evaluation settings

are different. Nevertheless, the comparison provides a reference point for assessing the effectiveness of the proposed lightweight architecture against an independently reproduced ECG classification framework.

TABLE IV
COMPARISON WITH IMLE-NET ON THE PTB-XL DATASET WITH OUR MODEL ON PHYSIONET/CINC 2020 DATASET.

Model	Macro AUROC	Fmax
IMLE-Net [2]	0.9216	0.8057
Our Reproduction	0.9189	0.4718

This difference in F1 scores should be considered in the context of the experimental settings and dataset difference. The original IMLE-Net evaluation [2] considers five diagnostic classes on PTB-XL [18], whereas the proposed framework uses 24 diagnostic classes on CinC2020. The larger number of classes and the different class distributions make the classification task more complex and prevent a direct comparison of the F1 values across the two datasets. Therefore, the IMLE-Net results are used primarily as a reproducibility reference and to provide contextual evidence for the performance of ECG classification architectures rather than as a direct benchmark for the proposed student model.

F. Comparison with Existing Self-Supervised ECG Models

To evaluate the effectiveness of the proposed lightweight student model in the context of self-supervised ECG representation learning, we compare it with the downstream classification results reported by Mehari and Strodthoff [19]. Their study systematically evaluated Contrastive Predictive Coding (CPC), SimCLR, and Bootstrap Your Own Latent (BYOL) using 12-lead ECG recordings and reported Macro AUC values of 0.9418, 0.9265, and 0.9290, respectively, on the PTB-XL downstream classification task.

For a more consistent comparison, the proposed student was separately fine-tuned and evaluated on PTB-XL using the same 71-class diagnostic setting. The student achieved a Macro AUROC of 0.8713 with only 107,239 parameters. The student does not perform self-supervised pretraining independently; instead, it receives knowledge through distillation from the proposed self-supervised pretrained teacher. Thus, it represents a knowledge-distilled lightweight model derived from self-supervised representations.

TABLE V
COMPARISON OF THE PROPOSED LIGHTWEIGHT STUDENT WITH EXISTING SELF-SUPERVISED ECG MODELS ON PTB-XL.

Model	SSL Method	Parameters	Macro AUROC
Mehari et al.	CPC	5.8M	0.9418
Mehari et al.	SimCLR	0.93M	0.9265
Mehari et al.	BYOL	0.93M	0.9290
Ours – Student	SSL + KD	0.107M	0.8713

The proposed student does not achieve the highest Macro AUROC among the compared models; however, it operates with substantially fewer parameters. With 107,239 parameters, the student is approximately $8.7\times$ smaller than the 0.93M-parameter SimCLR and BYOL models and approximately $54\times$ smaller than the 5.8M-parameter CPC model. The lower AUROC therefore reflects a trade-off between predictive performance and model compactness. The result demonstrates that the proposed knowledge distillation strategy can retain useful representations from the self-supervised teacher in a highly lightweight architecture, making the student more suitable for resource-constrained deployment.

IV. CONCLUSION

This work presented SSL-KD, a self-supervised teacher–student framework for lightweight multi-label ECG classification. The proposed framework combines self-supervised representation learning on unlabeled MIMIC-IV-ECG recordings with supervised teacher fine-tuning and knowledge distillation on the PhysioNet/CinC 2020 dataset. The teacher achieved a CM Score of 0.6527 and a Macro AUROC of 0.9486, while the 104,834-parameter student achieved a CM Score of 0.5378 and a Macro AUROC of 0.8949. Despite the substantial reduction in model size, the student retained useful diagnostic performance and achieved a Micro F1 of 0.7220.

The comparison with reproduced baseline models shows that the proposed student achieves competitive performance while using a substantially smaller architecture. The comparison with IMLE-Net further provides a reproducibility-based reference, although the different datasets and numbers of diagnostic classes prevent direct quantitative comparison. Overall, the results demonstrate the potential of combining self-supervised learning and knowledge distillation to develop compact ECG classification models for resource-constrained and edge-based healthcare applications.

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